```
# ----------------------------
# BEST BUY RETAIL STORE - POS SYSTEM
# ----------------------------

TAX_RATE = 0.10        # Sales tax rate (10%)
DISCOUNT_RATE = 0.05    # Discount rate (5%)
DISCOUNT_LIMIT = 5000   # Minimum amount to get discount
choice = ""           # Stores menu choice
item = ""             # Stores product name
qty = 0               # Stores quantity
price = 0.0           # Stores item price
stock = 0             # Stores stock amount
subtotal = 0.0        # Stores subtotal
tax = 0.0             # Stores tax amount
total = 0.0           # Stores total bill
payment = 0.0         # Stores customer payment
change = 0.0          # Stores change

# Product catalog stored in a dictionary
# Each product has a price and stock amount
# Value = [price , stock]
products = {
    "rice": [450.00, 20],       # 1kg rice
    "flour": [380.00, 15],      # 1kg flour
    "sugar": [360.00, 18],      # 1kg sugar
    "oil": [950.00, 10],        # 1L cooking oil
    "milk": [280.00, 25],       # 1L milk
    "eggs": [720.00, 12],       # dozen eggs
```

```
    "bread": [320.00, 20],      # loaf of bread
    "soap": [150.00, 30],       # bar soap
    "toothpaste": [420.00, 15],  # toothpaste
    "water": [120.00, 40]       # bottled water
}

# Empty dictionary to store cart items
cart = {}

# ----------------------------
# FUNCTION TO SHOW PRODUCTS
# ----------------------------
def show_products():
    # Print heading
    print("\nAvailable Products:")
    print("--------------------")

    # Loop through all products
    for item in products:
        # Get price and stock for each product
        price = products[item][0]
        stock = products[item][1]

        # Display product details
        print(item.capitalize(), "- $", price, "| Stock:", stock)

# ----------------------------
# FUNCTION TO ADD ITEM TO CART
```

```
# ----------------------------
def add_to_cart():
    # Show available products
    show_products()

    # Ask user for product name
    item = input("\nEnter product name: ").lower()

    # Check if product exists
    if item not in products:
        print("❌ Product not found.")
        return

    # Ask user for quantity
    qty = int(input("Enter quantity: "))

    # Check if enough stock is available
    if qty > products[item][1]:
        print("❌ Not enough stock.")
        return

    # Add item to cart or update quantity
    cart[item] = cart.get(item, 0) + qty

    # Reduce stock from product list
    products[item][1] -= qty

    # Confirm item added
```

```
    print("✅ Item added to cart.")

# ----------------------------
# FUNCTION TO REMOVE ITEM
# ----------------------------
def remove_from_cart():
    # Check if cart is empty
    if not cart:
        print("🛒 Cart is empty.")
        return

    # Show cart contents
    print("\nCart Items:")
    for item in cart:
        print(item.capitalize(), "- Quantity:", cart[item])

    # Ask user for item name
    item = input("Enter item to remove: ").lower()

    # Check if item exists in cart
    if item not in cart:
        print("❌ Item not in cart.")
        return

    # Ask quantity to remove
    qty = int(input("Enter quantity to remove: "))
```

```
    if qty >= cart[item]:
        # Return stock
        products[item][1] += cart[item]
        # Remove item from cart
        del cart[item]
    else:
        # Reduce quantity
        cart[item] -= qty
        # Return stock
        products[item][1] += qty

    # Confirm removal
    print("✅ Item removed.")

# ----------------------------
# FUNCTION TO VIEW CART
# ----------------------------
def view_cart():
    # Check if cart is empty
    if not cart:
        print("🛒 Cart is empty.")
        return

    # Initialize subtotal
    subtotal = 0

    discount = 0
```

```
# Print heading
print("\nYour Cart:")
print("-----------")

# Loop through cart items
for item in cart:
    price = products[item][0]
    qty = cart[item]
    total = price * qty

    # Add to subtotal
    subtotal += total

    # Show item details
    print(item.capitalize(), "-", qty, "x $", price, "= $", round(total, 2))

# Apply discount if subtotal exceeds limit
if subtotal > DISCOUNT_LIMIT:
    discount = subtotal * DISCOUNT_RATE

# Calculate tax on discounted subtotal
tax = (subtotal - discount) * TAX_RATE

# Calculate final total
total_bill = subtotal - discount + tax

# Display totals
print("\nSubtotal: $", round(subtotal, 2))
```

```
    print("Discount:", round(discount, 2))
    print("Tax (10%): $", round(tax, 2))
    print("Total: $", round(total_bill, 2))

# ----------------------------
# FUNCTION TO CHECKOUT + RECEIPT
# ----------------------------
def checkout():
    # Check if cart is empty
    if not cart:
        print("🛒 Cart is empty.")
        return

    # Show cart summary
    view_cart()

    # Calculate subtotal
    subtotal = 0
    for item in cart:
        subtotal += products[item][0] * cart[item]

    # Initialize discount
    discount = 0

    # Apply discount if eligible
    if subtotal > DISCOUNT_LIMIT:
        discount = subtotal * DISCOUNT_RATE
```

```
# Calculate tax
tax = (subtotal - discount) * TAX_RATE

# Calculate final total
total = subtotal - discount + tax

# Ask for payment
payment = float(input("\nEnter amount received: $"))

# Validate payment
if payment < total:
    print("❌ Payment not enough.")
    return

# Calculate change
change = payment - total

# Print receipt
print("\n--- RECEIPT ---")
print("Best Buy Retail Store")

# Print purchased items
for item in cart:
    print(item, "-", cart[item])

# Print totals
print("Subtotal:", subtotal)
print("Discount:", discount)
```

```
    print("Tax:", tax)
    print("Total:", total)
    print("Paid:", payment)
    print("Change:", change)

    # Thank customer
    print("Thank you for shopping!")

    # Clear cart for next transaction
    cart.clear()

# ----------------------------
# MAIN MENU LOOP
# ----------------------------
while True:
    # Display menu options
    print("\n--- POS MENU ---")
    print("1. View Products")
    print("2. Add Item to Cart")
    print("3. Remove Item from Cart")
    print("4. View Cart")
    print("5. Checkout")
    print("6. Exit")

    # Get user choice
    choice = input("Choose an option: ")

    # Menu options
```

```
if choice == "1":
    show_products()
elif choice == "2":
    add_to_cart()
elif choice == "3":
    remove_from_cart()
elif choice == "4":
    view_cart()
elif choice == "5":
    checkout()
elif choice == "6":
    print("Goodbye!")
    break
else:
    print("❌ Invalid option.")
```